

TECHNICAL DOCUMENTATION – Handy & HandyChat

ZJNU_HACKATHON_2025_TEAM10

1. Introduction

Handy & HandyChat is an AI-powered assistant designed to support international students at Zhejiang Normal University (ZJNU).

It centralizes scattered information into one accessible platform, helping students quickly find:

- PE class locations and schedules
- Sports facilities and teacher assignments
- Rules and regulations from the student handbook
- Answers to bilingual queries
- Combined questions (e.g., PE + graduation, PE + visa, etc.)

The project consists of:

- **Handy** — a lightweight, user-friendly web interface
- **HandyChat** — a backend chatbot system powered by GPT-4o-mini, MySQL, and a structured JSON-based handbook index

The system modernizes student support, reduces confusion, and lowers repetitive work for staff.

2. Project Objectives

- Provide instant, accurate answers to common student questions
- Offer bilingual (English/Chinese) support
- Eliminate the need for students to physically walk around campus for PE class information
- Reduce repetitive administrative questions handled by staff and volunteers
- Centralize PE and handbook information
- Build a scalable digital assistant that can grow over time

3. System Overview

Handy & HandyChat has two main components:

3.1 Handy (Frontend)

Developed using:

- **HTML** (templates folder)
- **CSS** (static/css)
- **Vanilla JavaScript** (static/js)

Provides:

- PE running rules
- Sports facility info + photos
- FAQ section
- Direct access to HandyChat
- Links and guidance for students (including Shining Team info)

3.2 HandyChat (Backend)

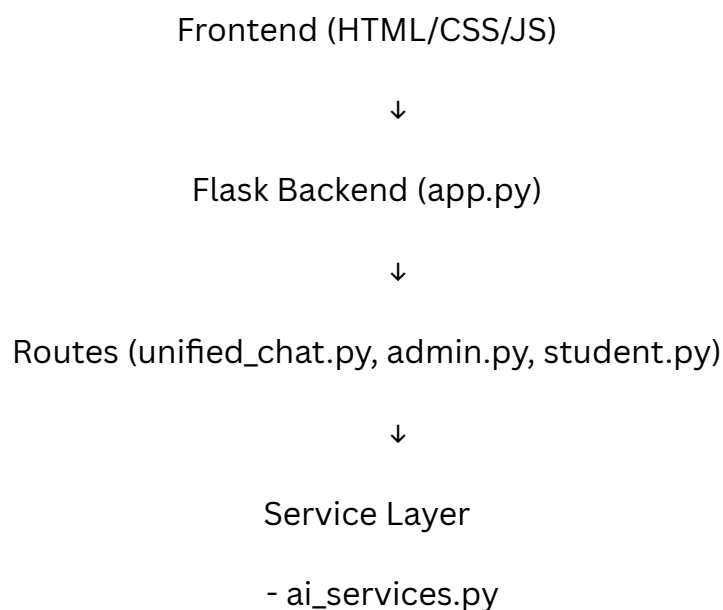
Built using:

- **Flask (Python)**
- **OpenAI GPT-4o-mini**
- **MySQL**
- **JSON + PDF handbook search**

Responsibilities:

- Receives student questions
- Detects intent (PE / Handbook / Combined)
- Retrieves PE class information from MySQL
- Searches handbook data using JSON & PDF fallback
- Combines results
- Enhances response using AI
- Returns structured JSON responses to the frontend

4. System Architecture



- database_services.py

- json_handbook_service.py

- handbook_search_service.py

- handbook_pdf_service.py

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MySQL (PE Data)

JSON Handbook (Handbook Content)

PDF Handbook (Backup Search via OCR)

This modular design keeps each responsibility isolated and easier to maintain.

5. Technologies & Tools Used (Accurate)

Backend

- Python 3.13
- Flask
- SQLAlchemy
- PyMySQL
- OpenAI API
- python-dotenv
- pdfplumber

Frontend

- HTML
- CSS
- JavaScript

Database

- MySQL

Development Tools

- VS Code
- Git & GitHub
- Git Bash
- Python test scripts
 - test_api.py
 - test_mysql.py
 - test_json_handbook.py

- test_endpoints.py
- test_flask_connection.py

6. Project Structure

(Updated to match your actual folder layout)

ZJNU_HACKATHON_2025_TEAM10/

```
└── HandyChat/
    ├── app.py
    ├── config.py
    ├── extensions.py
    ├── models.py
    ├── create_sample_data.py
    ├── migrate_handbook.py
    ├── requirements.txt
    ├── .env
    |
    ├── routes/
    |   ├── __init__.py
    |   ├── admin.py
    |   ├── student.py
    |   └── unified_chat.py
    |
    ├── services/
    |   ├── __init__.py
    |   ├── ai_services.py
    |   ├── database_services.py
    |   └── handbook_pdf_service.py
```

- | |—— handbook_search_service.py

- | |—— json_handbook_service.py

- |

- |—— static/

- | |—— css/

- | |—— handbook/

- | |—— images/

- | |—— js/

- | |—— manifest.json

- |

- |—— templates/

- | |—— chat.html

- | |—— index.html

- | |—— signup.html

- | |—— student.html

- |

- |—— Testing/

- | |—— debug_server.py

- | |—— interactive_test.ps1

- | |—— reset_database.py

- | |—— simple_chat.py

- | |—— test_api.py

- | |—— test_endpoints.py

- | |—— test_flask_connection.py

- | |—— test_json_handbook.py

- | |—— test_mysql.py

7. Database Design

✓ Location

Stores PE facility information.

✓ Teacher

Stores instructor information.

✓ SportClass

Stores sport class + foreign keys to Location & Teacher.

✓ ClassSchedule

Stores weekly schedule information for each sport class.

8. Core Functionalities (Technical)

8.1 Chat Handling

The main /ask endpoint:

1. Receives message
2. Sends to AI analyzer
3. Routes to correct services
4. Combines results
5. Returns final answer

8.2 PE Class Lookup

Uses:

- database_services.py
- SportClass, Location, Teacher, ClassSchedule

Returns:

- Location
- Schedule
- Teacher
- Class description
- Images

8.3 Handbook Lookup

Uses:

- json_handbook_service.py → main search
- handbook_search_service.py → scoring
- handbook_pdf_service.py → fallback scan

8.4 Bilingual Support

OpenAI classifies language and reformats responses.

9. API Endpoints (Summary)

POST /api/chat/ask

Main chatbot endpoint.

POST /api/chat/analyze

Intent-only analysis.

GET /api/chat/sports

Returns sports list.

GET /api/chat/health

Health check.

10. Testing Overview

You use pure Python test scripts:

- test_api.py → chat endpoint test
- test_mysql.py → DB connection test
- test_json_handbook.py → JSON search test
- test_endpoints.py → route tests
- test_flask_connection.py → app startup test

11. Environment & Deployment

Environment Variables (.env)

- OPENAI_API_KEY
- MYSQL_USER
- MYSQL_PASSWORD
- MYSQL_DB
- MYSQL_HOST

Run Steps

1. Install dependencies
2. Create database
3. Run create_sample_data.py or reset_database.py
4. Start app.py
5. Visit <http://localhost:5000>

12. Future Technical Improvements

- Add caching for handbook search
- Add admin CMS to update PE classes and handbook content
- Use JWT for student authentication
- Add unit test coverage for search ranking

13. Conclusion

This Technical Documentation provides a complete overview of the internal workings of Handy & HandyChat.

It defines the architecture, services, database, API structure, folder layout, and the technical flow that powers the system.